Data-based modeling of slug flow crystallization with uncertainty quantification

1st Anantha Prajith Kumar
Matr. 260192

2nd Giriraj Gautam
Matr. 259244

3rd Nabeel Zahid
Matr. 260322

4th Sabareesh Ramadoss Kolam
Matr. 260191

5th Dhanush Anand Shiva Balasubramani
Matr. 260193

***Abstract*- Accurate modeling of crystallization dynamics is essential for ensuring product quality in continuous pharmaceutical processes. This study presents a data-driven framework for modeling slug flow crystallization systems using artificial neural networks (ANNs) and quantifying predictive uncertainty through conformalized quantile regression (CQR). The dynamic behavior of the system is captured using a Nonlinear AutoRegressive architecture with eXogenous inputs (NARX), trained on sensor data from a slug flow crystallizer. To address uncertainty in the model's predictions, we implement a two-step CQR procedure that combines quantile regression with a residual-based calibration approach. Additionally, unsupervised learning techniques are used to identify different crystallization regimes within the dataset, ensuring that model training is process-specific. The framework is tested on real experimental data, demonstrating its ability to deliver reliable multi-step predictions and adaptive confidence intervals. The results show that the proposed approach offers both predictive accuracy and reliable uncertainty bounds, making it suitable for use in real-time process monitoring and control.**

Indexed terms- Slug flow crystallization, artificial neural networks, conformalized quantile regression, nonlinear autoregressive with eXogenous inputs

I. Introduction

Modeling and control of complex chemical processes, such as continuous crystallization, play a pivotal role in achieving consistent product quality and process reliability, especially in the pharmaceutical industry. Among various crystallization technologies, slug flow crystallization (SFC) offers promising benefits, including enhanced mixing, narrow residence time distribution, and better control over crystal size distribution. However, the dynamic and multi-phase nature of SFC poses significant challenges for first-principles modeling due to its nonlinear behavior, phase interactions, and sensor noise.

In recent years, data-driven modeling using machine learning techniques has emerged as a viable alternative to traditional physical models, particularly in scenarios where mechanistic modeling is either too complex or computationally expensive. Artificial neural networks (ANNs), when structured with temporal feedback mechanisms such as the Nonlinear AutoRegressive model with eXogenous inputs (NARX), are capable of learning the underlying dynamics of time-dependent systems directly from process data. These models are especially suitable for applications involving real-time control and prediction, as they can approximate complex system behavior without requiring explicit physical equations.

Despite the modeling capabilities of ANNs, one of their key limitations is the lack of interpretability in terms of uncertainty. Most neural networks provide point predictions without any notion of how confident the model is in those predictions. This is particularly problematic in critical applications like pharmaceutical crystallization, where uncertainty in model output must be accounted for to ensure robust process control and regulatory compliance.

To bridge this gap, this work integrates Conformalized Quantile Regression (CQR) into the ANN-based modeling pipeline. CQR enables the construction of valid prediction intervals that adapt to the heteroscedastic nature of the data—without requiring strong distributional assumptions. The combined approach allows not only the prediction of system behavior over time but also the quantification of uncertainty around those predictions, thereby supporting informed decision-making in the control of crystallization processes.

In this study, we develop a complete modeling and uncertainty quantification framework for a slug flow crystallizer using real process data. The workflow includes unsupervised clustering of different crystallization regimes, NARX-ANN-based dynamic modeling for each regime, and CQR-based construction of prediction intervals. This approach is tested on experimental datasets from a pharmaceutical crystallization process and evaluated in terms of prediction accuracy and uncertainty coverage.

II. Slug Flow Crystallization

Continuous pharmaceutical crystallization, especially in slug flow crystallizers (SFCs), offers improved product quality and process consistency. These reactor segments flow into alternating slugs of liquid and gas, providing enhanced control over mixing and crystal growth. However, their nonlinear dynamics, coupled with sensor noise and multiphase interactions, make physical modeling difficult and computationally intensive.

To overcome these challenges, data-driven modeling has emerged as a practical alternative. Nonlinear AutoRegressive models with eXogenous inputs (NARX), implemented using artificial neural networks (ANNs), are particularly suitable for capturing system dynamics from time-series data. These models leverage past system states and control inputs to predict future behavior without requiring detailed mechanistic equations.

The dataset used in this work consists of 98 experimental time-series files obtained from the operation of an SFC system. Each file includes synchronized measurements of key process variables such as temperatures (T_{TM}, T_{PM}), product concentration (c_{PM}), the full particle size distribution nout(L) is represented by its quantiles (d_{10}, d_{50}, d_{90}), and input flow rates (MF_{TM}, MF_{PM}, Q_{air} as Q_g). Additional features include the solid fraction in the feed ($\mathbf{w_{cryst}}$) and

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inlet conditions (c_{in}, $T_{PM,in}$, $T_{TM,in}$). These variables form the input and output vectors for model training and evaluation.

While ANN-based NARX models can learn complex dynamic behavior, they typically produce point estimates without conveying the uncertainty of their predictions. In pharmaceutical applications, this lack of confidence information is a significant limitation, especially for decision-making in real-time control.

III. Related Work

Continuous crystallization processes, especially slug flow systems, present a unique set of challenges for modeling and control due to their inherent nonlinearities and sensitivity to operating conditions. Traditional mechanistic models, while detailed, often require solving complex partial differential equations, which limits their practical use in real-time applications.

Johnson et al. [1] address this challenge by proposing a model predictive control (MPC) framework tailored for continuous crystallization. Their work emphasizes the use of surrogate models to approximate population balance equations, enabling faster computation. Although effective for control, their approach does not incorporate uncertainty quantification, which is crucial in pharmaceutical processes.

Termühlen et al. [2] provide a detailed experimental investigation of slug flow crystallization under varying operating conditions. Their findings highlight the impact of flow and design parameters on crystal size distribution, motivating the need for dynamic models that can capture these variations. However, their work remains experimental and does not extend to predictive modeling or control.

Romano et al. [3] introduce Conformalized Quantile Regression (CQR), a distribution-free method for generating statistically valid prediction intervals. CQR combines quantile regression with a calibration step to ensure finite-sample coverage guarantees. This makes it particularly valuable in settings with heteroscedastic noise and uncertain dynamics—characteristics typical of crystallization systems.

In aerospace and control domains, similar challenges in modeling complex dynamics have led to the adoption of surrogate models with embedded uncertainty. A recent study [4] explores the integration of neural networks with uncertainty quantification techniques for robust control, demonstrating the importance of interpretable predictions in safety-critical applications.

Building upon these contributions, our work integrates NARX-based neural network modeling with CQR to simultaneously achieve dynamic prediction and uncertainty quantification for slug flow crystallization. Unlike prior studies that focus solely on control or experimental analysis, we present a unified framework that models system behavior across multiple regimes and provides reliable prediction intervals for process monitoring and decision support.

IV. Methodology

This section outlines the data-driven framework developed to model the dynamics of a slug flow crystallizer and quantify prediction uncertainty. The procedure consists of five main stages: data preprocessing, regime clustering, lag-based input construction, dynamic modeling using a NARX-inspired neural network for identified processes, and uncertainty quantification through CQR.

A. Data preprocessing and Feature engineering

Each experimental run is provided as a time-series dataset containing key process measurements such as $\mathbf{T_{TM}}$, $\mathbf{T_{PM}}$, $\mathbf{c_{PM}}$, and crystal size metrics ($\mathbf{d_{10}}$, $\mathbf{d_{50}}$, $\mathbf{d_{90}}$). Prior to modeling, each trajectory is subjected to a structured data cleaning routine. Outlier detection is performed using the Interquartile Range (IQR) method, where any value falling outside 1.5 times the IQR beyond the first (Q1) or third quartile (Q3) is flagged as an anomaly. These points are then corrected using linear interpolation based on neighboring valid values instead of discarding them, thus preserving both the trend and temporal alignment of the signal.

After handling outliers, the dataset is normalized to bring all variables into a consistent scale and, where necessary, resampled to ensure uniform time steps across files. This preprocessing pipeline ensures that the model is trained on smooth, continuous, and dynamically coherent data, minimizing the influence of noise and preserving important signal patterns for learning.

B. Clustering of Trajectories for Regime Identification

To account for variability in crystallization behavior across experiments, a clustering step is performed to segment data into distinct process regimes. Feature vectors are extracted from each experimental run by aggregating input and output variables over time. These vectors, representing the overall dynamic profile of each trajectory, are used as input to a KMeans clustering algorithm. The optimal number of clusters (k) is selected using both the Elbow method and Silhouette score. To interpret the clustering results and assess regime separation visually, Principal Component Analysis (PCA) is applied post hoc to project the high-dimensional feature space into two principal components. Each resulting cluster corresponds to a distinct operating regime, enabling the development of specialized models that are better tuned to local process behavior and variation.

C. Lag-Based Input Construction

Slug-flow crystallization exhibits inherent transport delays (e.g., mixing, heat transfer, and crystal growth kinetics) that cause current control actions to influence measurable outputs only after several time steps. To capture these dynamics, the model is not asked to predict the *next* sample but rather the state at time $(k+g)$, where **g** is a tunable *prediction horizon* (also called the lag gap). For every prediction, the network receives a window of past information that embodies the process memory as:

- Input vector: $u\square = [u_k, u_{k-1}, ..., u_{k-n}]$
- Output history: $y\square = [y_{k-1}, y_{k-2}, ..., y_{k-n}]$

Here, n_u and n_y represents the number of lag steps for inputs and outputs, respectively.Including both past outputs (auto-regression) and past inputs (exogenous signals) transforms the network into a NARX-style predictor that can

learn closed-loop dynamics and actuator–response delays. It is important because,

- **Captures system delays:** Lagged inputs help the model learn how crystal size and concentration respond slowly to changes.
- **Reduces noise sensitivity:** Historical context helps smooth out sensor fluctuations.
- **Enables forward planning:** Predicting g steps ahead supports real-time control and optimization.
- **Improves rollout stability:** Training with a horizon reduces error drift in multi-step forecasts.

The final feature set includes process temperatures (T_{TM}, T_{PM}), product concentration (c_{PM}), crystal size quantiles (d_{10}, d_{50}, d_{90}), flow-related parameters (Q_g, MF_{TM}, MF_{PM}), and feed conditions such as cin, $T_{PM,in}$, $T_{TM,in}$, and w_{cryst}.

D. Dynamic Modeling Using Neural Networks

To capture the nonlinear and delayed-response dynamics of slug flow crystallization, a hybrid neural network architecture is employed. The model adopts a NARX-inspired structure by utilizing time-lagged sequences of both input variables (such as flow rates, feed concentrations, and temperatures) and output variables (like crystal size and solute concentration). This combination allows the model to learn causal relationships and system memory across time. To further enhance its ability to represent long-term temporal dependencies, the architecture incorporates Long Short-Term Memory (LSTM) layers. These recurrent components are well-suited for sequence modeling, enabling the network to capture non-linearities and delayed effects that traditional feedforward structures may miss.

Rather than training a single model across all data, regime-specific networks are trained based on the clustering of dynamic behavior identified during preprocessing. Each cluster reflects a distinct crystallization process. During inference, the test trajectory is first assigned to its most similar cluster using the same feature logic applied during training. The corresponding model is then used for prediction, ensuring the network operates within its domain of learned behavior. All models are trained using mean squared error (MSE) loss, with dropout and early stopping to prevent overfitting. This modeling strategy results in a robust forecasting system that is both temporally aware and structurally adaptive to different operating conditions.

E. Quantile Regression for Uncertainty Modeling

To estimate predictive uncertainty, two separate neural networks are trained using asymmetric pinball loss functions with q=0.05 and q=0.95 to predict the conditional quantiles of each output variable. The pinball loss penalizes underestimation and overestimation differently. These quantile-specific networks, trained on identical input features but optimized for different probability levels, provide adaptive prediction intervals that capture both aleatoric uncertainty from process noise and epistemic uncertainty from model limitations under varying operational conditions.

F. Conformalized Quantile Calibration

While quantile regression captures distributional uncertainty, its intervals may not meet the desired coverage levels. To address this, conformalized quantile regression (CQR) is used. A calibration set is held out to compute conformity scores based on the deviation of actual values from estimated quantiles. In the final implementation, a smoothed distribution of residuals is used to stabilize interval width and mitigate overfitting to noise. The final prediction interval for a new input is expanded using a quantile of these calibration errors, ensuring statistically valid and robust coverage.

$$\mathbf{E_i = \max \{ q_{0.05}(x_i) - y_i \, , \, y_i - q_{0.95}(x_i) \}}$$

Here, $q_{0.05}(x_i)$ and $q_{0.95}(x_i)$ are the predicted lower and upper quantiles, and y_i is the true observed value.

To construct the final interval for a new input x, the original quantile predictions are expanded by a correction factor (denoted $\hat{q}$), computed from the empirical distribution of the conformity scores in the calibration set. The final prediction interval becomes:

$$\mathbf{\hat{C}(x) = [\, q_{0.05}(x) - \hat{q} \, , \, q_{0.95}(x) + \hat{q} \,]}$$

This correction ensures that the interval contains the true value with the desired confidence level (e.g., 90%), regardless of the data distribution or model complexity. By incorporating CQR, the framework produces not only accurate predictions but also trustworthy and adaptive uncertainty estimates, which are essential for robust process.

V. Results

The modeling framework was evaluated on time-series data from a slug flow crystallization process. The goal was to assess both the accuracy of dynamic predictions and the reliability of the associated uncertainty estimates. The key outcomes of the study are discussed below.

A. Model Performance

The trained NARX-based neural network demonstrated strong predictive performance across different operational regimes. Using MSE as the evaluation metric, the model was able to forecast key state variables such as crystal size and process temperature with high accuracy. This validates the model's ability to learn the underlying system dynamics using historical input-output sequences and the selected lag structure.

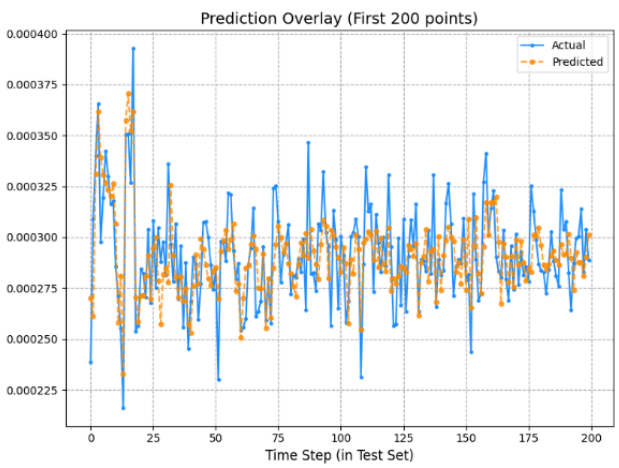

Fig. 1. Predicted vs. actual values of `d90` over the full test trajectory, showing good temporal alignment of model predictions with true system behavior.

B. Effectiveness of Regime Clustering

Clustering of experimental trajectories revealed meaningful regime-specific patterns in crystallization dynamics. The identified clusters corresponded to distinct process behaviors, such as variations in thermal profiles or product evolution. By assigning each test trajectory to its nearest cluster and training separate models per regime, the framework ensured that predictions were made using the most context-appropriate model. This structure-sensitive approach validated the utility of clustering, both in isolating dominant behavioral modes and in enhancing predictive accuracy.

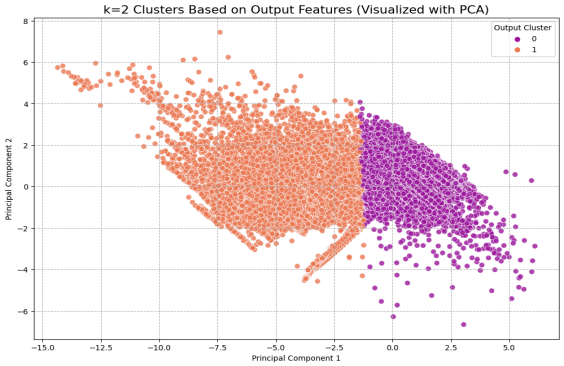

Fig. 2. Unsupervised Clustering with PCA

C. Uncertainty Quantification

Quantile regression models were successfully trained to estimate the 5th and 95th percentiles of the prediction distribution. These quantiles were used to construct preliminary prediction intervals. However, their raw coverage did not always match the target confidence level due to the variance in residuals.

To further enhance reliability, residuals used for conformal calibration were smoothed, reducing local variance and ensuring that the prediction intervals remained stable across high-frequency variations. The final results demonstrate that the majority of actual values fell within the 90% intervals, validating the effectiveness of the approach.

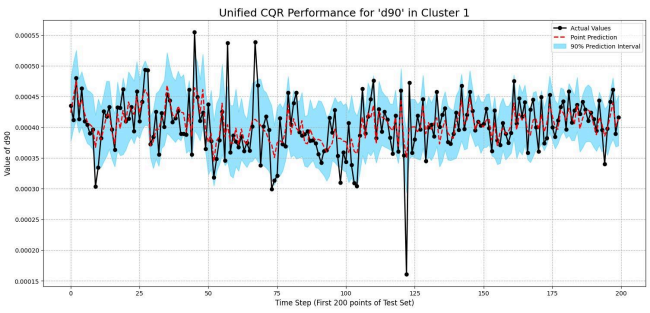

Fig. 3. 90% CQR prediction interval for `d90` in Cluster 1

D. Summary of Findings

The results support the following key conclusions:

- dynamic behavior of the crystallization process can be effectively modeled using a NARX-style neural architecture.
- incorporating regime-specific clustering leads to more accurate models.
- the combination of quantile regression and conformal calibration yields valid and interpretable prediction intervals as depicted in Fig. 4, making the framework suitable for risk-aware control strategies.

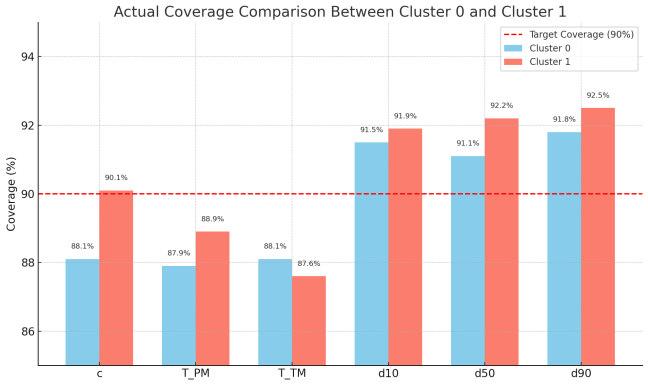

Fig. 4. Coverage of each features for both Clusters

Conclusion

This work concludes a data-driven modeling framework for dynamic prediction and uncertainty quantification in a continuous slug flow crystallization system. By leveraging experimental time-series data, a NARX-inspired neural network architecture was developed to forecast key process variables such as temperature, concentration, and crystal size distributions across varying operating regimes.

To address variability in process behavior, a clustering approach was used to group trajectories into regime-specific clusters, enabling specialized model training and improved generalization. The incorporation of lag-based input design and a prediction horizon allowed the model to capture temporal dependencies effectively.

Beyond accurate predictions, the framework also provides interpretable uncertainty estimates using conformalized quantile regression (CQR). This technique produced statistically valid prediction intervals with guaranteed coverage, as confirmed through calibration and visual validation. The resulting intervals support confidence-aware forecasting and pave the way for robust decision-making in model-based control strategies.

Overall, the integration of dynamic modeling with uncertainty quantification demonstrates strong potential for real-time monitoring and control of nonlinear, multiphase crystallization systems. Future work may extend this framework to include feedback control, adaptive modeling, and transfer learning across reactor scales or compounds.

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